

Proof techniques

↳ Properties of integers.

Well Ordering Principle

$$\textcircled{1} \mathbb{Z}^+ = \{x \in \mathbb{Z} \mid x > 0\} = \{x \in \mathbb{Z} \mid x \geq 1\}.$$

$$\textcircled{2} \mathbb{Q}^+ = \{x \in \mathbb{Q} \mid x > 0\} = ? \quad \times \quad \left| \begin{array}{l} \text{No precise} \\ \text{"Least"} \\ \text{element.} \end{array} \right.$$

$$\textcircled{3} \mathbb{R}^+ = \{x \in \mathbb{R} \mid x > 0\} = ? \quad \times$$

→ Any subset of $\mathbb{Z}^+$ (including itself) will have a LEAST ELEMENT.

This property of $\mathbb{Z}^+$ and its subsets having a least element is called the Well-Ordering Principle (WOP).

① ~~Find~~ Finite Induction Principle
or Principle of Mathematical
Induction.

Theorem 1:

- ① $S(n)$ denote an open statement.
 - ② n is a variable & a positive integer.
 - a) If $S(1)$ is true, and
 - b) If whenever $S(k)$ is true (for some some particular $k \in \mathbb{Z}^+$) then $S(k+1)$ is true,
- then $S(n)$ is true for $\forall n \in \mathbb{Z}^+$

Prove theorem 1.

Proof of theorem 1:

Let $S(n)$ be an open statement satisfying the conditions (a) & (b)

$$F = \{t \in \mathbb{Z}^+ \mid S(t) \text{ is false}\}.$$

We wish to prove: $F = \emptyset$
We assume that $F \neq \emptyset$ & prove by way of contradiction.

By the WOP, F must have a least element

$$\Rightarrow s$$

Now ~~s~~ $S(1)$ is true.

$$\therefore s \neq 1.$$

But if $s > 1$, then ~~s~~ $s-1 \in \mathbb{Z}^+$

$s-1 \notin F$ since then s cannot be the least element of F .

~~$S(s-1)$~~ $S(s-1)$ should be true.

Taking (b) from the principle: $S((s-1)+1)$ should be true. $\Rightarrow S(s)$ is true. $\Rightarrow$ Contradiction

$$\therefore F = \emptyset$$

□

Eg. 1. For any $n \in \mathbb{Z}^+$ prove that

$$\sum_{i=1}^n i = 1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

Base case : $S(1) = \text{true}$.

Inductive hypothesis $\sum_{i=1}^k i = \frac{k(k+1)}{2}$

Inductive step : $\sum_{i=1}^{k+1} i$

$$= \sum_{i=1}^k i + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1)$$

$$= \frac{(k+1)(k+2)}{2}$$

□.

E.g. 2.

Sum of odd numbers is always a perfect square.

$$\begin{array}{rcl} S(1) : 1 & = & 1^2 \\ \hline 1+3 & = & 2^2 \\ 1+3+5 & = & 3^2 \\ & \vdots & \\ & \vdots & \end{array}$$

Prove:

$$\sum_{i=1}^n 2i-1 = n^2$$

Inductive hypothesis:

$$\sum_{i=1}^k 2i-1 = k^2.$$

Inductive step:

$$\begin{aligned} \sum_{i=1}^{k+1} 2i-1 &= \sum_{i=1}^k 2i-1 + [2(k+1)-1] \\ &= k^2 + 2(k+1)-1 \\ &= k^2 + 2k+1 = (k+1)^2 \end{aligned}$$

□

E.g.3. Harmonic numbers:

$$H_1 = 1.$$

$$H_2 = 1 + \frac{1}{2}$$

$$H_3 = 1 + \frac{1}{2} + \frac{1}{3}$$

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}, \quad n \in \mathbb{Z}^+$$

Prove that:

$$\boxed{\text{If } n \in \mathbb{Z}^+, \quad \sum_{j=1}^n H_j = (n+1)H_n - n.}$$

Base:

$$\sum_{j=1}^1 H_j = H_1 = 1 = 2 \cdot 1 - 1 = (1+1)H_1 - 1.$$

Inductive hypothesis:

$$\sum_{j=1}^k H_j = (k+1)H_k - k.$$

Inductive step:

$$\sum_{j=1}^{k+1} H_j = \sum_{j=1}^k H_j + H_{k+1}$$

$$= [(k+1)H_k - k] + H_{k+1}$$

$$= (k+1)H_k - k + H_{k+1}$$

$$= (k+1) \left[H_{k+1} - \frac{1}{k+1} \right] - k + H_{k+1}$$

$$= (k+2) H_{k+1} - (k+1) \times \frac{1}{k+1} - k.$$

$$= (k+2) H_{k+1} - 1 - k$$

$$= (k+2) H_{k+1} - (k+1).$$

□.

Strong Mathematical Induction.

$S(n)$ denote an open statement where n is variable s.t. $n \in \mathbb{Z}^+$

Also let $n_0, n_1 \in \mathbb{Z}^+$, $n_0 \leq n_1$.

(a) If $S(n_0), S(n_0+1) \dots S(n_1-1)$ and $S(n_1)$ are true, and

(b) If whenever $S(n_0), S(n_0+1) \dots S(k-1)$ and $S(k)$ are true for $k \in \mathbb{Z}^+$ & $k \geq n_1$, then $S(k+1)$ is also true.

then $S(n)$ is true for all $n \geq n_0$.

E.g 4.

$$a_0 = 1, a_1 = 2, a_2 = 3.$$

$$a_n = a_{n-1} + a_{n-2} + a_{n-3} \quad \text{for all } n \in \mathbb{Z}^+ \text{ s.t. } n \geq 3.$$

We claim that if this recurrence holds then $a_n \leq 3^n$. Prove the claimed statement.

$$\begin{aligned} \text{Base: } a_0 &= 1 = 3^0 \leq 3^0 \\ a_1 &= 2 \leq 3 = 3^1 \\ a_2 &= 3 \leq 3^2 = 9 \end{aligned}$$

Holds for: $s(0), s(1), s(2)$

I. Hypothesis: The statements $s(0) \dots s(k-1), s(k)$ are all true.

$$\begin{aligned} \text{II. Step: } a_{k+1} &= a_k + a_{k-1} + a_{k-2} \\ &\leq 3^k + 3^{k-1} + 3^{k-2} \\ &\leq 3^k + 3^k + 3^k \\ &= 3(3^k) = 3^{k+1} \end{aligned}$$

□.

E.g 5.

$$F_0 = 0, F_1 = 1, F_2 = 1.$$

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \in \mathbb{Z}^+, n \geq 2.$$

H.W. p.t. (a) $\forall n \in \mathbb{Z}^+ \quad \sum_{i=0}^n F_i^2 = F_n \times F_{n+1}$

(b)

$$F_n \leq \left(\frac{5}{3}\right)^n.$$